

Evaluation Scenario Descriptions

Silver Ground Truth Dataset for the Group-Based Academic Project Recommender System

Generated from `Experimental_Groups_with_Recommendations.json`

36 Evaluation Scenarios · 107 Students · 36 Configuration Types

This document presents the full set of evaluation scenarios derived from the Silver Ground Truth dataset used to assess the group-based academic project recommender system. The recommender applies a balanced scoring strategy in which 50% of the recommendation weight is assigned to student competency (derived from course history and grades) and 50% to expressed preferences (interests and application domain priorities). Each scenario describes a synthetic but realistic student group, analyses its academic and interest profile, and evaluates the alignment between the group profile and the generated recommendations.

Scenario Summary Table

Table 1: Summary of Evaluation Scenarios

Group	Member	Competency Level	Interest Pattern	Application Orientation	RDIA	Top Recommended Theme
G001	3	Moderate	Overlapping	Healthcare / Medical; Education	Health & Wellness	Deep Learning for Medical Diagnosis
G002	3	Strong	Overlapping	Business & Productivity; Education	Economies of the Future	Web and Intelligent Productivity Platforms
G003	2	Strong	Overlapping	Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	Cybersecurity Threat Detection
G004	3	Strong	Overlapping	Sustainability & Environment; Transportation & Mobility	Sustainable Environment	IoT and Smart Physical Systems
G005	3	Moderate	Overlapping	Healthcare / Medical; Education	Health & Wellness	Computer Vision for Healthcare and Accessibility
G006	3	Moderate	Overlapping	Business & Productivity; Finance & Economics	Economies of the Future	Intelligent Recommender and Data Systems
G007	3	Moderate	Overlapping	Education; Social Good & Community	Health & Wellness	Natural Language Processing for Social Good
G008	3	Strong	Overlapping	Transportation & Mobility; Healthcare / Medical	Energy & Industrial Leadership	Robotics and Tracking for Healthcare

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Group	Member	Competency Level	Interest Pattern	Application Pat- tation	Orien- RDIA	Top Recommended Theme
G009	3	Moderate	Overlapping	Security & Privacy; Fi- nance & Economics	Energy & Industrial Leadership	Network Security and Threat Intelligence
G010	3	Strong	Overlapping	Healthcare / Medical; Education	Health & Wellness	AI and Bioinformatics for Healthcare
G011	3	Moderate	Overlapping	Sustainability & Envi- ronment; Business & Productivity	Sustainable Environ- ment	IoT Sensing and Smart Infrastructure
G012	3	Strong	Overlapping	Education; Entertain- ment & Media	Economies of the Fu- ture	Web Applications for Education and Enter- tainment
G013	3	Moderate	Overlapping	Healthcare / Medical; Education	Health & Wellness	Deep Learning for Medical Imaging
G014	3	Strong	Overlapping	Business & Productiv- ity; Finance & Eco- nomics	Economies of the Fu- ture	Web and Mobile Busi- ness Applications
G015	3	Moderate	Overlapping	Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	Cybersecurity and Network Forensics
G016	3	Strong	Overlapping	Transportation & Mo- bility; Sustainability & Environment	Energy & Industrial Leadership	Autonomous Robotics and AI Systems

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Group	Member	Competency Level	Interest Pattern	Application Pat- tation	Orien- RDIA	Top Recommended Theme
G017	3	Moderate	Overlapping	Healthcare / Medical; Education	Health & Wellness	AI and NLP for Healthcare and Education
G018	3	Strong	Overlapping	Business & Productivity; Education	Economies of the Future	Interactive Web and Game Applications
G019	4	Moderate	Overlapping	Sustainability & Environment; Healthcare / Medical	Sustainable Environment	IoT Security and Smart Monitoring
G020	3	Strong	Overlapping	Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	Cybersecurity and Secure Software Engineering
G021	3	Moderate	Overlapping	Healthcare / Medical; Transportation & Mobility	Health & Wellness	AI Vision for Healthcare and Transport
G022	3	Strong	Overlapping	Business & Productivity; Entertainment & Media	Economies of the Future	Web and Entertainment Platforms
G023	3	Moderate	Overlapping	Sustainability & Environment; Healthcare / Medical	Energy & Industrial Leadership	Robotics and IoT for Smart Systems
G024	3	Strong	Overlapping	Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	Network Security and Threat Intelligence

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Group	Member	Competency Level	Interest Pattern	Pat- tation	Application	Orien- RDIA	Top Recommended Theme
G025	3	Moderate	Diverse		Business & Productivity; Education	Economies of the Future	E-Commerce and Business Web Systems
G026	2	Strong	Diverse		Healthcare / Medical; Education	Health & Wellness	AI and Bioinformatics for Health
G027	3	Moderate	Diverse		Sustainability & Environment; Business & Productivity	Sustainable Environment	IoT and Environmental Smart Monitoring
G028	2	Strong	Diverse		Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	AI-Powered Cybersecurity Detection
G029	3	Moderate	Diverse		Security & Privacy; Government & Public Sector	Energy & Industrial Leadership	Network Security and Access Control
G030	3	Strong	Diverse		Healthcare / Medical; Education	Sustainable Environment	AI-Vision for Healthcare and Social Impact
G031	4	Moderate	Diverse		Education; Social Good & Community	Economies of the Future	Interactive Educational Web Platforms
G032	3	Strong	Diverse		Security & Privacy; Healthcare / Medical	Energy & Industrial Leadership	AI-Driven Security and IoT Protection

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Group	Member	Competency Level	Interest Pattern	Application	Orien- RDIA	Top Recommended Theme
G033	3	Moderate	Diverse	Healthcare / Medical; Transportation & Mobility	Health & Wellness	Computer Vision and Autonomous Systems
G034	3	Strong	Diverse	Security & Privacy; Healthcare / Medical	Energy & Industrial Leadership	AI-Assisted Cybersecurity Systems
G035	3	Moderate	Diverse	Sustainability & Environment; Transportation & Mobility	Sustainable Environment	IoT and Robotics for Smart Mobility
G036	3	Strong	Diverse	Security & Privacy; Healthcare / Medical	Energy & Industrial Leadership	AI and Security for Diverse Domains

Scenario Case Descriptions

The following subsections present structured scenario descriptions for all 36 evaluation groups. Each description follows a standardised format comprising five analytical components: (1) Group Profile, (2) Interest and Application Profile, (3) Recommendation Outcome, (4) Alignment Analysis, and (5) Expected Project Direction.

Scenario 1 — AI-Driven Medical Imaging and Diagnostic Support

Group: G001 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Computer Vision, Data Science

Top Applications: Healthcare / Medical, Education, Finance & Economics

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G001 consists of three students whose coursework is concentrated in deep learning, image processing, and natural language processing. The group achieved an average GPA equivalent of approximately 3.40 across all enrolled courses, reflecting a moderate-to-strong academic performance. The academic orientation of the group is firmly rooted in applied artificial intelligence, with relevant prior study in neural networks, digital image processing, and advanced AI electives.

Interest and Application Profile

The group's dominant interest is Artificial Intelligence and Machine Learning, complemented by Computer Vision and Data Science. These interests are largely shared across members, creating a cohesive technical focus. Application-wise, the group prioritises the Healthcare and Medical domain as its primary target, followed by Education and Finance. This profile indicates a strong preference for deploying intelligent systems in high-impact societal sectors.

Recommendation Outcome

The top-ranked recommendation is *COVID-19 Detection Using Deep Neural Networks*, followed by computer vision-based assistive systems, bioinformatics projects, and AI-enabled rehabilitation tools. The dominant theme across the top-10 results is the application of deep learning and computer vision to biomedical and accessibility challenges. Projects such as deepfake detection and crowd counting in public spaces also reflect the group's breadth in visual AI.

Alignment Analysis

The recommendations demonstrate strong alignment with the group's profile. The competency weight correctly elevates projects requiring deep neural network expertise, which matches the group's coursework in CS1465 and CS1464. The interest weight reinforces healthcare-oriented outcomes, producing a coherent and well-matched recommendation list. The RDIA category of Health & Wellness is consistently reflected in the top results.

Expected Project Direction

This group is most likely to pursue a deep learning project in the medical imaging domain, such as disease detection from radiological scans, AI-assisted diagnosis, or a computer vision system for assistive healthcare technology.

Scenario 2 — Web-Based Intelligent Applications for Business and Education

Group: G002 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Economies of the Future

Top Interests: Web Development, Cloud Computing, Mobile Development

Top Applications: Business & Productivity, Education, Entertainment & Media

Top Recommended Project: *CCIS Student Assistant Chatbot*

Group Profile

Group G002 comprises three students with a strong academic background in web application development, mobile development, and database systems. The group achieved a high average GPA equivalent of approximately 3.65, placing it in the Strong competency tier. The students have concentrated their coursework in platform-based application development and software engineering, reflecting readiness for end-to-end software product development.

Interest and Application Profile

The group's interests converge on Web Development, Cloud Computing, and Mobile Development. All three members share an interest in web development, making this the anchoring technical focus. Application domains prioritised by the group include Business and Productivity, Education, and Entertainment. This orientation reflects a preference for user-facing, commercially deployable digital products.

Recommendation Outcome

The top-ranked recommendation is *CCIS Student Assistant Chatbot*, reflecting the intersection of web interfaces and intelligent automation. Other highly ranked projects include e-commerce categorisation systems, faculty rating platforms, and web-based audio summarisation tools. The dominant theme across the top-10 list is the development of intelligent web-based platforms serving educational and business productivity needs.

Alignment Analysis

The recommendations are well-aligned with the group's technical and preference profile. The competency-based scoring favours projects requiring software engineering and application development skills, while the interest and application weights surface productivity and education-oriented systems. The RDIA category of Economies of the Future is consistently represented, underscoring the group's orientation toward digital transformation.

Expected Project Direction

This group would most likely develop a full-stack web or mobile application targeting educational or business productivity use cases, potentially incorporating lightweight AI features such as recommendation logic or chatbot interfaces.

Scenario 3 — Machine Learning-Based Cyber Threat Detection

Group: G003 · **Members:** 2 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Network Systems, Database Systems

Top Applications: Security & Privacy, Government & Public Sector, Finance & Economics

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G003 is a two-student group with strong academic performance (average GPA equivalent of 3.72) in cybersecurity-related coursework, including network security and digital forensics. Despite its small size, the group demonstrates focused technical depth in security topics, making it one of the higher-performing groups in the dataset.

Interest and Application Profile

Both members share a primary interest in Cybersecurity, with complementary interests in Network Systems and Database Systems. The application focus is on Security and Privacy, with secondary interest in Government and Public Sector and Finance. The group exhibits a sharply defined security-first profile, with applications concentrated in high-trust institutional environments.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, followed by botnet detection, adversarial malware detection, and deepfake video forensics. Machine learning-assisted threat detection constitutes the dominant theme. Projects on security visualisation tools, video surveillance, and cyberbullying detection round out the top-10 list.

Alignment Analysis

The alignment is strong and precise. The competency weight correctly surfaces security-engineering projects aligned with the students' coursework in CS1474 and CS1476. The interest weight further reinforces cyber-focused outcomes. The RDIA category of Energy and Industrial Leadership reflects a national priority in infrastructure protection, consistent with the security application domain.

Expected Project Direction

This group is likely to pursue a machine learning-powered cybersecurity project, such as an automated malware classification system, a network intrusion detection engine, or a tool for identifying adversarial attacks in deployed models.

Scenario 4 — Embedded and IoT Solutions for Sustainable Smart Environments

Group: G004 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Sustainable Environment

Top Interests: Internet of Things (IoT), Embedded Systems, Robotics

Top Applications: Sustainability & Environment, Transportation & Mobility, Healthcare / Medical

Top Recommended Project: *Smart Car Parking Manager (SCPM)*

Group Profile

Group G004 consists of three students with diverse but complementary backgrounds spanning robotics, embedded systems, IoT, and artificial intelligence. The group achieved a strong average GPA equivalent of 3.65. The academic profile reflects hands-on systems-level expertise, with coursework in embedded programming, IoT fundamentals, robotics, and AI.

Interest and Application Profile

The group's shared interest is Internet of Things (IoT), supported by individual interests in Embedded Systems, Robotics, and AI/ML. This creates a technically cohesive profile oriented toward physical computing and smart systems. Application priorities include Sustainability and the Environment, Transportation, and Healthcare, reflecting an orientation toward impactful real-world deployments of connected devices.

Recommendation Outcome

The top recommendation is *Smart Car Parking Manager (SCPM)*, indicative of the system's recognition of the group's strengths in IoT and real-time embedded control. Additional recommendations include GPS tracking systems, environmental monitoring platforms, and smart infrastructure management tools, consistently aligned with the group's sustainability and mobility interests.

Alignment Analysis

The recommendations demonstrate excellent alignment. The competency scoring elevates projects requiring embedded and IoT engineering proficiency, while preference scoring focuses the output on sustainability and transport domains. The RDIA category of Sustainable Environment and Supply of Essential Needs is directly reflected in the recommended project themes.

Expected Project Direction

This group is best suited for an IoT-based smart system project, such as a real-time environmental monitoring platform, a connected transportation management system, or an embedded sensor network for sustainability applications.

Scenario 5 — Computer Vision Applications for Medical and Assistive Contexts

Group: G005 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Computer Vision, Artificial Intelligence / Machine Learning, Data Science

Top Applications: Healthcare / Medical, Education, Transportation & Mobility

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G005 comprises three students whose academic backgrounds are concentrated in neural networks, digital image processing, and AI. The group's average GPA equivalent of 3.41 places it in the Moderate competency tier, reflecting consistent but not exceptional academic performance. The shared academic exposure to deep learning and visual data processing defines the group's technical base.

Interest and Application Profile

The dominant shared interest is Computer Vision, complemented by Artificial Intelligence/Machine Learning and Data Science. Application domains prioritised include Healthcare/Medical, Education, and Transportation. A subset of students also exhibits interest in Robotics, broadening the group's potential project scope into intelligent autonomous systems.

Recommendation Outcome

The top recommendation mirrors G001 and G013 in surfacing *COVID-19 Detection Using Deep Neural Networks* as the primary result, reflecting the recommender's consistent recognition of AI-healthcare alignment. Additional top results include computer vision systems for accessibility, autonomous navigation, and educational handwriting recognition.

Alignment Analysis

The alignment is strong with respect to both competency and interest dimensions. However, the application match is only partial, as Transportation-oriented projects are somewhat underrepresented in the top results relative to the group's stated interest. Healthcare dominates, which is consistent but may not fully reflect the breadth of the group's application preferences.

Expected Project Direction

This group would most likely develop a computer vision project for clinical or assistive applications, potentially extending into autonomous navigation or smart transport systems depending on the specific member interests pursued.

Scenario 6 — Distributed Data Management and Context-Aware Recommendation

Group: G006 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Economies of the Future

Top Interests: Distributed Systems, Cloud Computing, Database Systems

Top Applications: Business & Productivity, Finance & Economics, Education

Top Recommended Project: *Online Context-Aware Recommender System*

Group Profile

Group G006 consists of three students with a moderate competency profile (average GPA equivalent of 3.37), specialising in distributed systems, cloud computing, and databases. The coursework background reflects preparedness for large-scale data infrastructure and intelligent system design, though the academic performance is the lowest among the Moderate tier groups.

Interest and Application Profile

The shared interest is Distributed Systems, complemented by Cloud Computing, Database Systems, and Mobile Development. Application priorities are Business and Productivity, Finance, and Education. The group's profile reflects a preference for scalable back-end systems and intelligent data services rather than end-user interface development.

Recommendation Outcome

The top recommendation is *Online Context-Aware Recommender System*, reflecting the recommender's detection of the group's affinity for data-driven, back-end-oriented software. Other top results include collaborative filtering systems, database management tools, and intelligent web services, all consistent with the group's systems-level orientation.

Alignment Analysis

Alignment is moderate-to-strong. The competency scoring correctly identifies data engineering and distributed systems competencies, while the preference scoring surfaces business and productivity applications. The Economies of the Future RDIA is well-represented, though Finance-specific projects are underrepresented relative to stated application priorities.

Expected Project Direction

This group is well-positioned to develop a data-driven back-end system, such as a context-aware recommendation engine, a distributed data pipeline, or a cloud-hosted database

service for business analytics.

Scenario 7 — NLP-Based Systems for Education and Community Services

Group: G007 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Natural Language Processing, Artificial Intelligence / Machine Learning, Data Science

Top Applications: Education, Social Good & Community, Finance & Economics

Top Recommended Project: *Examining the effectiveness of pretrained language models for Arabic NLP tasks*

Group Profile

Group G007 is a three-student group with a moderate competency level (average GPA 3.41), primarily skilled in natural language processing, AI, and data science. The academic profile reflects exposure to NLP theory and applied machine learning, with supporting coursework in optimisation and advanced AI topics.

Interest and Application Profile

The anchoring shared interest is Natural Language Processing, with complementary interests in AI/ML and Data Science. Application priorities span Education, Social Good and Community, and Finance. This profile indicates a preference for language-based AI applications serving educational and civic purposes.

Recommendation Outcome

The top recommendation is *Examining the Effectiveness of Pre-trained Language Models for Arabic NLP Tasks*, directly reflecting the group's NLP focus and Arabic-language context. Other top results include sentiment analysis, text classification, dialogue systems, and Arabic text processing tools. The Education and Social Good application domains are well-represented.

Alignment Analysis

The alignment is strong and highly specific. The recommendation engine correctly identifies NLP as the dominant competency and interest, producing a coherent list of language technology projects. The RDIA of Health & Wellness appears less directly reflected in the NLP context, though some results (e.g., healthcare-related text mining) bridge this gap.

Expected Project Direction

This group is most likely to develop an Arabic NLP system, such as a text classification model, a sentiment analysis tool for social media, or a language model fine-tuning experiment for domain-specific tasks.

Scenario 8 — Autonomous Tracking and Robotics for Vulnerable Populations

Group: G008 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping · **RDIA:** Energy & Industrial Leadership

Top Interests: Robotics, Embedded Systems, Internet of Things (IoT)

Top Applications: Transportation & Mobility, Healthcare / Medical, Sustainability & Environment

Top Recommended Project: *Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs*

Group Profile

Group G008 consists of three students with a strong competency level (average GPA 3.65), combining expertise in robotics, embedded systems, and IoT. The group diverges slightly from G004 through one member's interest in Game Development, introducing a heterogeneous element into an otherwise technically focused profile.

Interest and Application Profile

The primary shared interest is Robotics, complemented by Embedded Systems, Internet of Things, and Game Development. Application domains prioritised include Transportation, Healthcare, and Sustainability. The group's profile suggests readiness for physically deployed intelligent systems, particularly in mobility and safety contexts.

Recommendation Outcome

The top recommendation is *Monitoring and Tracking of Vulnerable Individuals Using Miniature Robots*, directly aligning with the group's robotics expertise and healthcare application interest. Subsequent recommendations include surveillance drones, smart tracking systems, and embedded navigation applications, consistent with the group's transportation and safety orientation.

Alignment Analysis

The alignment is highly precise. Both the competency and interest scoring components converge on robotics and IoT engineering projects. The healthcare and transportation application domains are well-reflected. The RDIA of Energy and Industrial Leadership is partially aligned, as industrial robotics is a natural extension of the group’s technical profile.

Expected Project Direction

This group is best suited for a robotics-based project, such as an autonomous tracking system for healthcare environments, a miniature surveillance robot, or a smart mobility solution using embedded sensor technology.

Scenario 9 — Network Vulnerability Analysis and Security Intelligence

Group: G009 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Network Systems, Software Engineering

Top Applications: Security & Privacy, Finance & Economics, Government & Public Sector

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G009 is a three-student group with moderate academic performance (average GPA 3.41) specialising in cybersecurity and network systems. The academic background includes courses in network security, software security, and digital forensics, providing a solid but not exceptional foundation in the security domain.

Interest and Application Profile

The shared interest is Cybersecurity, supported by Network Systems, Software Engineering, and Database Systems. Application priorities are Security and Privacy, Finance, and Government. The group’s profile is characteristic of students preparing for security engineering or threat analysis roles in institutional or governmental contexts.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, consistent with other cybersecurity-focused groups in the dataset. The top-10 results

include network intrusion detection, malware analysis, visualisation tools for cryptographic algorithms, and forensic investigation systems. The security theme is highly concentrated across all recommended projects.

Alignment Analysis

The alignment is strong and consistent. Both scoring dimensions point toward network and software security projects. However, the Finance application domain is relatively underrepresented in the recommendations compared to the group's stated priorities. The RDIA of Energy and Industrial Leadership is indirectly addressed through the infrastructure protection theme.

Expected Project Direction

This group is most likely to pursue a network security project, such as an automated intrusion detection system, a security monitoring dashboard, or a tool for detecting and classifying network-level attacks.

Scenario 10 — AI-Enabled Bioinformatics and Clinical Decision Support

Group: G010 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Bioinformatics, Computer Vision

Top Applications: Healthcare / Medical, Education, Transportation & Mobility

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G010 comprises three students with strong academic performance (average GPA 3.65) in deep learning, computer vision, NLP, and bioinformatics-adjacent AI. The group's academic profile reflects a broad but coherent exposure to intelligent systems, with an emphasis on applying machine learning to complex data domains.

Interest and Application Profile

The shared interest is Artificial Intelligence/Machine Learning, with complementary interests in Bioinformatics, Computer Vision, and Natural Language Processing. Application domains prioritised include Healthcare, Education, and Transportation. The combination

of bioinformatics and clinical AI creates a distinctive research-oriented profile within the dataset.

Recommendation Outcome

The top recommendation is again *COVID-19 Detection Using Deep Neural Networks*, reflecting the Healthcare and AI intersection. Other highly ranked results include DNA analysis tools, bioinformatics sequence processing, and AI-based rehabilitation systems. The thematic emphasis on healthcare AI and molecular biology distinguishes this group's recommendation set.

Alignment Analysis

Alignment is strong. The bioinformatics interest, which is relatively rare in the dataset, is well-captured by the recommendation engine, elevating projects such as genetic region analysis. The competency in deep learning further reinforces the AI-heavy recommendation profile. The Health & Wellness RDIA is directly and consistently reflected.

Expected Project Direction

This group is best positioned to pursue a bioinformatics or clinical AI project, such as a genomic sequence analyser, a predictive model for disease onset, or a deep learning tool for medical image classification.

Scenario 11 — IoT-Based Monitoring Systems for Sustainable Infrastructure

Group: G011 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Sustainable Environment

Top Interests: Internet of Things (IoT), Embedded Systems, Distributed Systems

Top Applications: Sustainability & Environment, Business & Productivity, Government & Public Sector

Top Recommended Project: *Energy-aware Multi-Object Detection and Compression in WMSN*

Group Profile

Group G011 is a three-student group with moderate academic performance (average GPA 3.46), combining expertise in IoT, embedded systems, and distributed architectures. The academic profile reflects systems-level competency with a particular emphasis on sensor-driven and networked computing environments.

Interest and Application Profile

The shared interest is Internet of Things (IoT), supported by Embedded Systems, Distributed Systems, and Database Systems. Application priorities include Sustainability and the Environment, Business and Productivity, and Government. The group's profile reflects an orientation toward large-scale connected infrastructure, particularly in environmental management contexts.

Recommendation Outcome

The top recommendation is *Energy-aware Multi-Object Detection and Compression in Wireless Multimedia Sensor Networks (WMSN)*, directly addressing the group's IoT and distributed systems competencies. Other top results include smart environmental sensors, IoT-based logistics tracking, and distributed infrastructure management platforms.

Alignment Analysis

The alignment is strong and technically precise. Both competency and interest scoring contribute to the emergence of IoT and distributed sensing projects. The sustainability and government application domains are well-represented. The RDIA of Sustainable Environment and Supply of Essential Needs is directly reflected in the top recommendations.

Expected Project Direction

This group would most likely develop an IoT-based environmental monitoring or smart infrastructure system, potentially incorporating distributed data aggregation, energy optimisation, and real-time sensing components.

Scenario 12 — Platform Development for Educational and Digital Media

Group: G012 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping · **RDIA:** Economies of the Future

Top Interests: Web Development, Software Engineering, Game Development

Top Applications: Education, Entertainment & Media, Finance & Economics

Top Recommended Project: *CCIS Student Assistant Chatbot*

Group Profile

Group G012 consists of three students with strong academic performance (average GPA 3.65) in web development, game development, and distributed systems. The group's

academic profile reflects strong software engineering capabilities with a creative orientation toward interactive application development.

Interest and Application Profile

The anchoring shared interest is Web Development, supported by Software Engineering, Game Development, and Distributed Systems. Application domains prioritised include Education, Entertainment and Media, and Finance. This combination reflects a group that bridges technical software engineering with creative interactive media.

Recommendation Outcome

The top recommendation is *CCIS Student Assistant Chatbot*, reflecting the web development and educational application alignment. Other top results include game-based learning systems, interactive media platforms, and web-based productivity tools. The dual orientation toward education and entertainment is well-captured by the recommendation set.

Alignment Analysis

The alignment is moderate-to-strong. Web development and software engineering competencies are correctly identified, and the education and entertainment application domains are well-represented. However, game development projects are somewhat underrepresented in the top-10 results relative to the group's stated interest, possibly due to the limited number of game-specific projects in the project pool.

Expected Project Direction

This group would most likely develop a web-based educational platform or interactive media application, potentially incorporating gamification elements or game-driven learning components.

Scenario 13 — Deep Learning-Based Diagnostic Systems in Medical Imaging

Group: G013 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Computer Vision, Artificial Intelligence / Machine Learning, Data Science

Top Applications: Healthcare / Medical, Education, Business & Productivity

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G013 is a three-student group with moderate academic performance (average GPA 3.41) whose coursework spans neural networks, image processing, and advanced AI. The group shares two overlapping interests—Computer Vision and Data Science—making it one of the more cohesive groups in its interest configuration.

Interest and Application Profile

The group's dominant shared interests are Computer Vision and Data Science, supplemented by Artificial Intelligence/Machine Learning and Bioinformatics. Application domains prioritised include Healthcare/Medical, Education, and Business and Productivity. The overlap in both interests and applications creates a focused, well-defined group profile.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, consistent with other AI-healthcare groups. Subsequent recommendations include medical image classification systems, AI-powered accessibility tools, and data-driven educational applications. The recommendations capture the healthcare-first orientation with some representation of business analytics.

Alignment Analysis

The alignment is strong. The dual interest overlap in Computer Vision and Data Science, combined with the Healthcare application priority, produces a coherent and well-matched recommendation set. The Health & Wellness RDIA is consistently reflected. The Business and Productivity application domain is somewhat underrepresented.

Expected Project Direction

This group is most likely to pursue a computer vision or data science project with a medical imaging or clinical analytics focus, such as a diagnostic classification system or a health data visualisation dashboard.

Scenario 14 — Cross-Platform Application Development for Business Services

Group: G014 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Economies of the Future

Top Interests: Web Development, Mobile Development, Cloud Computing

Top Applications: Business & Productivity, Finance & Economics, Education

Top Recommended Project: *Qaimni: An Interactive, Web-Based System for Rating Faculty Members*

Group Profile

Group G014 comprises three students with strong academic performance (average GPA 3.65) in web, mobile, and distributed application development. The group shares interests in both Web Development and Mobile Development, creating a multi-platform software development profile oriented toward business services.

Interest and Application Profile

The group's overlapping interests in Web Development and Mobile Development are complemented by Cloud Computing and Distributed Systems. Application priorities include Business and Productivity, Finance, and Education. This profile characterises a group with full-stack development capabilities oriented toward delivering scalable digital services.

Recommendation Outcome

The top recommendation is *Qaimni: An Interactive, Web-Based System for Rating Faculty Members*, reflecting the group's web development strength and education application preference. Other top results include e-commerce platforms, mobile tracking systems, and cloud-integrated business applications. The business and education domains are well-captured.

Alignment Analysis

The alignment is strong across both scoring dimensions. The competency weight surfaces platform development projects consistent with the group's coursework, while the preference weight correctly orients results toward business and educational applications. The Economies of the Future RDIA is consistently and appropriately reflected.

Expected Project Direction

This group is best positioned to develop a full-stack web or mobile application serving business or educational contexts, such as a faculty management system, an e-commerce service platform, or a cloud-based productivity tool.

Scenario 15 — Integrated Cybersecurity and Network Forensics Systems

Group: G015 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Network Systems, Database Systems

Top Applications: Security & Privacy, Government & Public Sector, Finance & Economics

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G015 is a three-student group with moderate academic performance (average GPA 3.41) specialising in cybersecurity, network security, and database management. The group shares two overlapping interests—Cybersecurity and Database Systems—creating a distinctive profile that combines threat detection with data management expertise.

Interest and Application Profile

The dominant interests are Cybersecurity and Database Systems, supplemented by Network Systems and Distributed Systems. Application priorities include Security and Privacy, Government and Public Sector, and Finance. This profile reflects a group oriented toward institutional security infrastructure and data governance.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, consistent with other cybersecurity-focused groups. The top-10 list also includes network forensics tools, database security systems, and access control platforms. The Security and Government application domains are well-represented.

Alignment Analysis

The alignment is strong. Both competency and interest scoring converge on security and data management projects. The inclusion of database security-specific results distinguishes this group's recommendation set from other pure-network-security groups. The Energy and Industrial Leadership RDIA is partially addressed through critical infrastructure protection themes.

Expected Project Direction

This group would most likely pursue a cybersecurity project with a data management component, such as a secure database access control system, a network forensics and

anomaly detection platform, or a security audit tool for institutional data environments.

Scenario 16 — Robotics and Autonomous Systems for Industrial Applications

Group: G016 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Energy & Industrial Leadership

Top Interests: Robotics, Artificial Intelligence / Machine Learning, Embedded Systems

Top Applications: Transportation & Mobility, Sustainability & Environment, Healthcare / Medical

Top Recommended Project: *Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs*

Group Profile

Group G016 comprises three students with strong academic performance (average GPA 3.65) whose expertise spans robotics, AI, and embedded systems. The group's overlapping interests in Robotics and Embedded Systems, combined with strong AI coursework, define a technically advanced profile suited for autonomous system development.

Interest and Application Profile

The shared interests are Robotics and Embedded Systems, supplemented by Artificial Intelligence/Machine Learning and Natural Language Processing. Application domains prioritised include Transportation, Sustainability, and Healthcare. The group's profile reflects an orientation toward intelligent physical systems deployed in real-world mobility and environmental contexts.

Recommendation Outcome

The top recommendation is *Monitoring and Tracking of Vulnerable Individuals Using Miniature Robots*, reflecting the robotics and healthcare intersection. Other top results include autonomous navigation systems, smart mobility platforms, and AI-embedded sensor systems for environmental monitoring. The Transportation and Sustainability application domains are well-captured.

Alignment Analysis

The alignment is strong. The competency scoring correctly identifies robotics and embedded AI expertise, while the interest and application scoring reinforces autonomous systems

and sustainability projects. The Energy and Industrial Leadership RDIA aligns with the group's interest in intelligent physical infrastructure.

Expected Project Direction

This group is best suited for an autonomous robotics project in a transportation or environmental context, such as a self-navigating delivery robot, an AI-enabled environmental patrol drone, or an embedded smart vehicle system.

Scenario 17 — Multimodal AI Systems for Healthcare and Accessibility

Group: G017 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Computer Vision, Natural Language Processing

Top Applications: Healthcare / Medical, Education, Business & Productivity

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G017 is a three-student group with moderate academic performance (average GPA 3.41) that shares two overlapping interests: Artificial Intelligence/Machine Learning and Natural Language Processing. The group's coursework in deep learning, advanced AI, and NLP provides a strong theoretical and practical foundation in language-aware AI systems.

Interest and Application Profile

The anchoring shared interests are AI/ML and NLP, with Computer Vision as a complementary interest for one member. Application priorities include Healthcare/Medical, Education, and Business and Productivity. The multimodal interest profile—spanning language, vision, and data intelligence—reflects a versatile group capable of tackling complex AI tasks.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, placing the healthcare AI theme at the forefront. Other top results include NLP-based educational tools, AI-assisted accessibility systems, and visual recognition systems. The recommendations reflect the group's multimodal AI capabilities.

Alignment Analysis

The alignment is strong. The convergence of NLP and Computer Vision interests with a Healthcare application priority produces a recommendation set that is both technically coherent and application-relevant. The Health & Wellness RDIA is well-represented throughout the top-10 list.

Expected Project Direction

This group would most likely pursue a multimodal AI project combining language understanding and visual processing, such as an AI-based clinical documentation system, an NLP-enhanced diagnostic tool, or a vision-language model for accessibility applications.

Scenario 18 — Web and Game Development for Entertainment and Productivity

Group: G018 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Economies of the Future

Top Interests: Web Development, DevOps, Game Development

Top Applications: Business & Productivity, Education, Entertainment & Media

Top Recommended Project: *CCIS Student Assistant Chatbot*

Group Profile

Group G018 consists of three students with strong academic performance (average GPA 3.65) in web development, game development, and distributed systems. The group shares overlapping interests in Web Development and Game Development, creating a technically capable profile oriented toward interactive digital experiences.

Interest and Application Profile

The overlapping interests are Web Development and Game Development, supplemented by DevOps and Distributed Systems. Application priorities include Business and Productivity, Education, and Entertainment and Media. This profile indicates a group capable of building interactive, scalable web platforms with a focus on user engagement.

Recommendation Outcome

The top recommendation is *CCIS Student Assistant Chatbot*, reflecting the web development and education alignment. Other top results include game-integrated learning

tools, interactive entertainment platforms, and web-based productivity systems. The Entertainment and Education application domains are appropriately surfaced.

Alignment Analysis

The alignment is moderate. Web development projects are well-represented, and educational applications are consistently present. However, game development projects are somewhat underrepresented in the top results, a limitation likely attributable to the relative scarcity of game development entries in the underlying project pool. The Economies of the Future RDIA is consistently reflected.

Expected Project Direction

This group is best positioned to develop a game-integrated web application, an educational gamification platform, or an interactive productivity tool combining DevOps practices with engaging front-end experiences.

Scenario 19 — Security-Aware IoT Systems for Environmental Monitoring

Group: G019 · **Members:** 4 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Sustainable Environment

Top Interests: Internet of Things (IoT), Embedded Systems, Distributed Systems

Top Applications: Sustainability & Environment, Healthcare / Medical, Security & Privacy

Top Recommended Project: *Energy-aware Multi-Object Detection and Compression in WMSN*

Group Profile

Group G019 is the largest group in the dataset, comprising four students with a moderate competency level (average GPA 3.10). The group's academic background spans IoT, embedded systems, distributed systems, and cybersecurity, representing the most technically diverse configuration in the evaluation set.

Interest and Application Profile

The group shares three overlapping interests—IoT, Distributed Systems, and Cybersecurity—across its four members, reflecting a security-aware systems engineering orientation. Application priorities include Sustainability and the Environment, Healthcare, and Security

and Privacy. The breadth of interest coverage reflects the larger group size and diverse academic trajectories.

Recommendation Outcome

The top recommendation is *Energy-aware Multi-Object Detection and Compression in Wireless Multimedia Sensor Networks*, reflecting the IoT and distributed systems convergence. Other top results include secure IoT architectures, smart environmental monitoring systems, and cybersecurity tools for connected devices. The environmental and security dimensions are both well-represented.

Alignment Analysis

The alignment is moderate-to-strong. The overlapping interests in IoT, Distributed Systems, and Cybersecurity are all reflected in the top recommendations, though the breadth of the group's profile introduces some dispersion in results. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and consistently addressed.

Expected Project Direction

This group is best suited for a security-aware IoT project, such as a secure environmental monitoring system, a distributed IoT platform with intrusion detection capabilities, or a smart infrastructure management solution with built-in security layers.

Scenario 20 — Software Security and Cyber Threat Mitigation

Group: G020 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Software Engineering, Network Systems

Top Applications: Security & Privacy, Government & Public Sector, Finance & Economics

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G020 comprises three students with strong academic performance (average GPA 3.65) specialising in cybersecurity, software engineering, and network systems. The group shares overlapping interests in Cybersecurity and Software Engineering, representing a

profile focused on the intersection of secure software development and network-level threat management.

Interest and Application Profile

The shared interests are Cybersecurity and Software Engineering, complemented by Network Systems and Distributed Systems. Application priorities include Security and Privacy, Government and Public Sector, and Finance. This profile characterises a group oriented toward enterprise-grade security engineering and institutional protection.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, consistent with other cybersecurity-focused groups. The top-10 list also includes secure coding analysis tools, network monitoring systems, and software vulnerability detection platforms. The Security and Government application domains are well-reflected.

Alignment Analysis

The alignment is strong. The dual emphasis on Cybersecurity and Software Engineering is appropriately captured, surfacing projects that require both development and security competencies. The Energy and Industrial Leadership RDIA is partially addressed through critical software infrastructure themes.

Expected Project Direction

This group would most likely develop a secure software engineering tool, such as a static analysis platform for vulnerability detection, a secure DevSecOps pipeline, or an automated security testing framework.

Scenario 21 — AI and Vision-Based Systems for Healthcare and Smart Mobility

Group: G021 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Computer Vision, Data Science

Top Applications: Healthcare / Medical, Transportation & Mobility, Education

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G021 is a three-student group with moderate academic performance (average GPA 3.46) specialising in AI, computer vision, and NLP. The group shares overlapping interests in both AI/ML and Computer Vision, creating a visually oriented AI profile complemented by language processing capabilities.

Interest and Application Profile

The dominant shared interests are Artificial Intelligence/Machine Learning and Computer Vision, with Natural Language Processing and Data Science as complementary interests. Application domains include Healthcare/Medical, Transportation, and Education. The group's preference for Healthcare and Transportation suggests interest in deploying AI systems in high-stakes, real-world environments.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, consistent with the Healthcare AI theme observed across similar groups. Other top results include computer vision systems for assistive mobility, autonomous vehicle detection, and educational recognition tools. Transportation and healthcare are both well-captured.

Alignment Analysis

The alignment is strong. The shared Computer Vision and AI interests, combined with the Healthcare and Transportation application priorities, produce a well-matched set of recommendations. The Health & Wellness RDIA is directly reflected throughout the top results.

Expected Project Direction

This group is best suited for a computer vision project in a healthcare or mobility context, such as an AI-enabled diagnostic tool, an autonomous traffic monitoring system, or a vision-based navigation aid for mobility-impaired users.

Scenario 22 — Web-Based Platforms for Entertainment and Digital Commerce

Group: G022 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Economies of the Future

Top Interests: Web Development, Cloud Computing, Game Development

Top Applications: Business & Productivity, Entertainment & Media, Finance & Economics

Top Recommended Project: *CCIS Student Assistant Chatbot*

Group Profile

Group G022 comprises three students with strong academic performance (average GPA 3.60) in web, mobile, and game development. The group shares overlapping interests in Web Development and Game Development, reflecting a creative and technically capable profile oriented toward digital consumer applications.

Interest and Application Profile

The shared interests are Web Development and Game Development, complemented by Cloud Computing and Database Systems. Application domains prioritised include Business and Productivity, Entertainment and Media, and Finance. This profile reflects a group capable of delivering commercially viable, user-facing digital products.

Recommendation Outcome

The top recommendation is *CCIS Student Assistant Chatbot*, reflecting the web development and productivity alignment. Other top results include recommendation systems, e-commerce platforms, and interactive media applications. The Entertainment and Business application domains are appropriately surfaced.

Alignment Analysis

The alignment is moderate. Web development and business productivity projects are well-captured. Entertainment and game-related projects are somewhat underrepresented, likely due to the composition of the underlying project pool. The Economies of the Future RDIA is consistently reflected.

Expected Project Direction

This group would most likely develop a web-based entertainment or productivity application, potentially incorporating gamification, digital commerce features, or cloud-integrated interactive services.

Scenario 23 — Embedded Robotics and IoT for Sustainability and Healthcare

Group: G023 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Overlapping · **RDIA:** Energy & Industrial Leadership

Top Interests: Robotics, Internet of Things (IoT), Embedded Systems

Top Applications: Sustainability & Environment, Healthcare / Medical, Entertainment & Media

Top Recommended Project: *Monitoring and Tracking of Vulnerable Individuals Using Mini-UAVs*

Group Profile

Group G023 is a three-student group with moderate academic performance (average GPA 3.37) combining expertise in robotics, IoT, embedded systems, and computer graphics. The group shares overlapping interests in Robotics and Embedded Systems, with one member introducing a computer graphics interest that broadens the group’s creative technical scope.

Interest and Application Profile

The anchoring shared interests are Robotics and Embedded Systems, supplemented by Internet of Things and Computer Graphics. Application domains include Sustainability and the Environment, Healthcare, and Entertainment and Media. The inclusion of Entertainment as an application area—driven by the computer graphics interest—distinguishes this group from other robotics-focused groups.

Recommendation Outcome

The top recommendation is *Monitoring and Tracking of Vulnerable Individuals Using Miniature Robots*, consistent with the robotics and healthcare alignment. Other top results include smart environmental sensor systems, IoT-enabled visual displays, and embedded multimedia systems. The sustainability and healthcare themes dominate.

Alignment Analysis

The alignment is moderate. Robotics and IoT projects are well-represented, and the healthcare and sustainability application domains are consistently captured. The Entertainment application dimension and the computer graphics interest are underrepresented in the recommendations, reflecting the relative scarcity of graphics-specific projects in the available pool.

Expected Project Direction

This group is best positioned for a robotics or embedded systems project with environmental or healthcare applications, potentially incorporating visualisation components for monitoring dashboards or interactive data displays.

Scenario 24 — Network Security and Threat Detection Engineering

Group: G024 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Overlapping
· **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Software Engineering, Network Systems

Top Applications: Security & Privacy, Government & Public Sector, Business & Productivity

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G024 comprises three students with strong academic performance (average GPA 3.60) specialising in cybersecurity, network security, and distributed systems. The group shares overlapping interests in Cybersecurity and Network Systems, with Software Engineering and Distributed Systems as complementary interests.

Interest and Application Profile

The dominant shared interests are Cybersecurity and Network Systems, supported by Software Engineering and Distributed Systems. Application priorities include Security and Privacy, Government and Public Sector, and Business and Productivity. This profile reflects a group oriented toward enterprise network security and critical infrastructure protection.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, consistent with the broader cybersecurity cluster in the dataset. The top-10 results include network traffic analysis tools, intrusion detection systems, and security policy management platforms. The government and security application domains are well-represented.

Alignment Analysis

The alignment is strong. Both competency and interest scoring consistently elevate network security and threat intelligence projects. The Business and Productivity application dimension is less prominently reflected compared to its ranking among the group's stated priorities. The Energy and Industrial Leadership RDIA is indirectly addressed through critical network infrastructure themes.

Expected Project Direction

This group would most likely pursue a network security project, such as an intrusion detection and response system, a distributed firewall management tool, or a machine learning model for real-time network traffic classification.

Scenario 25 — Web and Database Systems for E-Commerce and Productivity

Group: G025 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Diverse · **RDIA:** Economies of the Future

Top Interests: Web Development, Cloud Computing, Database Systems

Top Applications: Business & Productivity, Education, Finance & Economics

Top Recommended Project: *Musanif: A Categorization and Product Information API for E-Commerce Product Images*)

Group Profile

Group G025 is a three-student group with moderate academic performance (average GPA 3.41) whose members exhibit a highly diverse set of interests spanning web development, cloud computing, database systems, mobile development, DevOps, and human-computer interaction. This group represents one of the widest interest spreads in the dataset with no overlapping interests between members.

Interest and Application Profile

The group has no shared interests, with each student contributing distinct technical priorities: Web Development and DevOps, Database Systems and Cloud Computing, and Mobile Development and Human-Computer Interaction. Application domains include Business and Productivity, Education, and Finance. Despite the interest diversity, the application orientation remains coherent and business-focused.

Recommendation Outcome

The top recommendation is *Musanif: A Categorisation and Product Information API for E-Commerce Product Images*, reflecting the group's aggregated profile across web, data, and business domains. Other top results include productivity systems, educational web applications, and database-driven services. The Business and Productivity theme dominates.

Alignment Analysis

The alignment is moderate. The diversity of interests challenges the recommender’s ability to converge on a single thematic outcome, resulting in a broader but less precise recommendation set. The aggregated application preference for Business and Productivity provides a coherent anchor. The Economies of the Future RDIA is consistently reflected.

Expected Project Direction

This group would benefit from a project that requires multiple technical roles, such as a full-stack e-commerce platform, a cloud-hosted business analytics service, or a productivity application integrating database back-ends with mobile front-ends.

Scenario 26 — AI-Enabled Medical and Bioinformatics Research Applications

Group: G026 · **Members:** 2 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Bioinformatics, Robotics

Top Applications: Healthcare / Medical, Education, Transportation & Mobility

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G026 is a two-student group with strong academic performance (average GPA 3.58) combining expertise in AI, bioinformatics, robotics, and embedded systems. With no overlapping interests between the two members, the group represents a complementary pairing of AI research and physical systems engineering.

Interest and Application Profile

The first member contributes interests in AI/ML and Bioinformatics, while the second contributes Robotics and Embedded Systems. Application priorities include Healthcare/Medical, Education, and Transportation. Despite the lack of interest overlap, the Healthcare application domain provides a unifying project direction.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, reflecting the Healthcare AI theme anchored by the first member’s profile. Other top results include bioinformatics tools and AI-assisted medical systems. Robotics and embedded systems-oriented results are less prominently represented.

Alignment Analysis

The alignment is partially strong and partially weak. The AI and bioinformatics member's profile is well-captured, but the robotics and embedded systems member's competencies are less reflected in the top results. The aggregated Healthcare application preference provides cohesion, but the recommendation set may not fully serve both members' capabilities equally.

Expected Project Direction

This group would be best served by a healthcare project that bridges AI and physical systems, such as a bioinformatics-informed robotic rehabilitation device, a smart wearable health monitor, or an AI-guided embedded diagnostic tool.

Scenario 27 — IoT-Driven Environmental Sensing and Smart Data Systems

Group: G027 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Diverse · **RDIA:** Sustainable Environment

Top Interests: Internet of Things (IoT), Cloud Computing, Distributed Systems

Top Applications: Sustainability & Environment, Business & Productivity, Finance & Economics

Top Recommended Project: *Energy-aware Multi-Object Detection and Compression in WMSN*

Group Profile

Group G027 consists of three students with moderate academic performance (average GPA 3.41) whose interests span IoT, cloud computing, distributed systems, database systems, NLP, and bioinformatics. With no overlapping interests, this group represents one of the most technically diverse configurations in the evaluation dataset.

Interest and Application Profile

The three members contribute distinct interest sets: IoT and Cloud Computing, Database Systems and Distributed Systems, and Natural Language Processing and Bioinformatics. Application priorities include Sustainability and the Environment, Business and Productivity, and Finance. The sustainability application domain provides a shared direction despite the interest diversity.

Recommendation Outcome

The top recommendation is *Energy-aware Multi-Object Detection and Compression in Wireless Multimedia Sensor Networks*, anchored by the IoT member's profile. Other top results include distributed data systems, smart environmental platforms, and NLP-informed data services. The sustainability theme is consistently present.

Alignment Analysis

The alignment is moderate. The IoT and sustainability connection is well-captured, while the NLP and bioinformatics interests are less directly reflected. The distributed systems and cloud computing competencies contribute to the systems-level recommendation outputs. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and consistently reflected.

Expected Project Direction

This group would most effectively pursue a multi-component project, such as an IoT-based environmental data platform with cloud back-end infrastructure and NLP-driven reporting or data interpretation features.

Scenario 28 — Intelligent Threat Detection with AI and Security Engineering

Group: G028 · **Members:** 2 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Artificial Intelligence / Machine Learning, Software Engineering

Top Applications: Security & Privacy, Government & Public Sector, Data Science

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G028 is a two-student group with strong academic performance (average GPA 3.65) combining expertise in cybersecurity, software engineering, AI, and data science. With no overlapping interests between the two members, this group represents a complementary security-AI pairing.

Interest and Application Profile

The first member contributes interests in Cybersecurity and Software Engineering, while the second contributes AI/ML and Data Science. Application priorities include Security

and Privacy, Government and Public Sector, and Data Science as an implied application. The security application domain is the primary unifying factor.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, which uniquely bridges both members' profiles by requiring both AI/ML methodology and cybersecurity domain knowledge. Other top results include adversarial attack detection, AI-driven forensics tools, and intelligent network threat classifiers.

Alignment Analysis

The alignment is strong and well-balanced across both members. The recommendation engine successfully identified the AI-cybersecurity intersection as the primary output domain, producing results that require contributions from both competency sets. The Energy and Industrial Leadership RDIA is appropriately reflected in the critical security infrastructure theme.

Expected Project Direction

This group is ideally suited for a machine learning-powered security project, such as an AI-based malware classifier, a predictive threat intelligence system, or an adversarial robustness evaluation framework for deployed models.

Scenario 29 — Comprehensive Network Security and Governance Systems

Group: G029 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Diverse · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Network Systems, Software Engineering

Top Applications: Security & Privacy, Government & Public Sector, Business & Productivity

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G029 is a three-student group with moderate academic performance (average GPA 3.41) specialising in cybersecurity, network systems, software engineering, cloud computing, databases, and distributed systems. With no overlapping interests, the group represents a broad security-infrastructure profile spanning multiple systems domains.

Interest and Application Profile

The three members contribute distinct interests: Cybersecurity and Network Systems, Software Engineering and Cloud Computing, and Database Systems and Distributed Systems. Application priorities include Security and Privacy, Government and Public Sector, and Business and Productivity. The security and governance application domains provide thematic coherence.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, consistent with the cybersecurity anchor in the group's aggregated profile. The top-10 results also include cloud security tools, secure database management systems, and network governance platforms. The government and security application domains are well-reflected.

Alignment Analysis

The alignment is moderate-to-strong. The cybersecurity and network systems competencies are well-captured in the top results. Cloud computing and distributed systems are less prominently represented, though back-end security and infrastructure management projects partially address these interests. The Energy and Industrial Leadership RDIA is indirectly addressed.

Expected Project Direction

This group would be best suited for a comprehensive security infrastructure project requiring multiple roles, such as a cloud-hosted network security monitoring system, a distributed access control platform, or a secure database management tool for government applications.

Scenario 30 — AI and Computer Vision for Clinical and Social Applications

Group: G030 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Sustainable Environment

Top Interests: Artificial Intelligence / Machine Learning, Computer Vision, Natural Language Processing

Top Applications: Healthcare / Medical, Education, Social Good & Community

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G030 comprises three students with strong academic performance (average GPA 3.65) whose diverse interests span AI/ML, computer vision, NLP, IoT, embedded systems, and data science. With no overlapping interests, this group represents the widest AI-driven interest diversity among the non-overlapping groups.

Interest and Application Profile

The three members contribute: AI/ML and Computer Vision, NLP and Data Science, and IoT and Embedded Systems. Application priorities include Healthcare/Medical, Education, and Social Good and Community. Despite the interest diversity, the healthcare and education application domains provide a coherent social impact orientation.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, anchored by the AI and healthcare convergence of the first member's profile. Other top results include NLP-based social applications, IoT-enabled health monitoring systems, and computer vision tools for accessibility. The social good and healthcare themes are consistently present.

Alignment Analysis

The alignment is moderate. The AI and healthcare dimensions are well-captured, while IoT and embedded systems competencies are less prominently represented in the top results. The Social Good and Community application domain is partially reflected in some NLP and accessibility-oriented recommendations. The RDIA of Sustainable Environment and Supply of Essential Needs is moderately represented.

Expected Project Direction

This group would benefit from a socially impactful AI project that integrates multiple technologies, such as a health monitoring system combining IoT sensors with AI-based analysis, or an NLP-powered community service platform with data science back-end components.

Scenario 31 — Human-Centered Web and Interactive Application Design

Group: G031 · **Members:** 4 · **Competency:** Moderate · **Interest Pattern:** Diverse ·
RDIA: Economies of the Future

Top Interests: Web Development, Human-Computer Interaction, Game Development

Top Applications: Education, Social Good & Community, Entertainment & Media

Top Recommended Project: *Qaimni: An Interactive, Web-Based System for Rating Faculty Members*

Group Profile

Group G031 is the second four-student group in the dataset, with moderate academic performance (average GPA 3.21). The group’s interests span web development, human-computer interaction, game development, computer graphics, distributed systems, database systems, cloud computing, and DevOps, representing the widest interest diversity in the entire dataset with no overlapping interests.

Interest and Application Profile

The four members contribute entirely distinct interest sets covering the full breadth of software engineering, interactive media, systems infrastructure, and user experience design. Application priorities include Education, Social Good and Community, Entertainment and Media, Business and Productivity, and Government and Public Sector. The Education and Social Good domains provide the primary unifying direction.

Recommendation Outcome

The top recommendation is *Qaimni: An Interactive, Web-Based System for Rating Faculty Members*, reflecting the intersection of web development, HCI, and educational applications. Other top results include interactive learning platforms, community-oriented applications, game-based education tools, and cloud-hosted educational services. The Education and Social Good themes are consistently well-represented.

Alignment Analysis

The alignment is moderate. The recommendation engine successfully identifies the education and social good application cluster as the coherent output domain, despite the extreme interest diversity. Individual interest areas such as DevOps, computer graphics, and cloud computing are only partially captured. The Economies of the Future RDIA is consistently reflected.

Expected Project Direction

This group is best suited for a large, multi-component educational or community platform project that leverages diverse technical roles, such as a cloud-deployed interactive learning system, a community service application with game elements, or a HCI-focused social impact platform.

Scenario 32 — Hybrid AI-Cybersecurity Solutions for Connected Environments

Group: G032 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Artificial Intelligence / Machine Learning, Internet of Things (IoT)

Top Applications: Security & Privacy, Healthcare / Medical, Sustainability & Environment

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G032 comprises three students with strong academic performance (average GPA 3.65) whose interests span cybersecurity, AI/ML, IoT, network systems, bioinformatics, and distributed systems. The group has no overlapping interests, representing a broad and technically diverse profile within a security-dominant context.

Interest and Application Profile

The three members contribute: Cybersecurity and Network Systems, AI/ML and Bioinformatics, and IoT and Distributed Systems. Application priorities include Security and Privacy, Healthcare/Medical, and Sustainability and the Environment. The security application domain is the primary unifying anchor despite the interest diversity.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, reflecting the cybersecurity and AI intersection that bridges two of the three members' profiles. Other top results include AI-driven security tools, IoT security frameworks, and intelligent infrastructure monitoring platforms. The security and healthcare themes are well-represented.

Alignment Analysis

The alignment is strong for the cybersecurity and AI dimensions but moderate for the IoT, bioinformatics, and sustainability dimensions. The recommendation engine successfully identifies the AI-cybersecurity intersection as the primary output domain. The Energy and Industrial Leadership RDIA is appropriately and consistently reflected.

Expected Project Direction

This group is best positioned for a hybrid AI-security project with optional IoT integration, such as an intelligent malware detection system for IoT devices, a bioinformatics-informed security risk model, or an AI-powered network monitoring platform with sustainability-aware optimisation.

Scenario 33 — Vision-Based AI for Healthcare and Autonomous Navigation

Group: G033 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Diverse · **RDIA:** Health & Wellness

Top Interests: Artificial Intelligence / Machine Learning, Computer Vision, Robotics

Top Applications: Healthcare / Medical, Transportation & Mobility, Business & Productivity

Top Recommended Project: *COVID-19 Detection Using Deep Neural Networks*

Group Profile

Group G033 is a three-student group with moderate academic performance (average GPA 3.41) whose diverse interests span AI/ML, computer vision, robotics, embedded systems, database systems, and cloud computing, with no overlapping interests between members. The group represents a broad AI and systems engineering profile.

Interest and Application Profile

The three members contribute: AI/ML and Computer Vision, Robotics and Embedded Systems, and Database Systems and Cloud Computing. Application priorities include Healthcare/Medical, Transportation, and Business and Productivity. Despite the interest diversity, the healthcare and transport application domains provide a coherent deployment context.

Recommendation Outcome

The top recommendation is *COVID-19 Detection Using Deep Neural Networks*, anchored by the healthcare AI profile. Other top results include autonomous navigation systems, embedded vision platforms, and cloud-hosted analytics tools. The healthcare and transportation themes are both present in the top-10 results.

Alignment Analysis

The alignment is moderate. Healthcare AI projects are well-captured for the first member, while robotics and embedded systems projects are partially addressed in the mid-tier recommendations. Database and cloud computing competencies are less prominently reflected. The Health & Wellness RDIA is consistently present.

Expected Project Direction

This group would be best suited for a multi-role project combining AI/CV for diagnosis with autonomous systems for delivery or transport, such as a smart robotic health assistant, an autonomous vehicle-mounted diagnostic sensor, or a cloud-integrated embedded vision system.

Scenario 34 — Intelligent Cybersecurity with AI and Web Integration

Group: G034 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Artificial Intelligence / Machine Learning, Web Development

Top Applications: Security & Privacy, Healthcare / Medical, Business & Productivity

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G034 comprises three students with strong academic performance (average GPA 3.60) whose interests span cybersecurity, AI/ML, web development, network systems, bioinformatics, and DevOps, with no overlapping interests. The group combines a security specialist, an AI researcher, and a web developer.

Interest and Application Profile

The three members contribute: Cybersecurity and Network Systems, AI/ML and Bioinformatics, and Web Development and DevOps. Application priorities include Security and Privacy, Healthcare/Medical, and Business and Productivity. The security and healthcare domains are the primary unifying application focuses.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, reflecting the cybersecurity and AI-ML intersection. Other top results include web-based

security dashboards, AI-driven forensics tools, and secure coding platforms. The security application theme dominates the top-10 list.

Alignment Analysis

The alignment is strong for the security and AI dimensions. The web development and DevOps interests are partially captured through security platform and dashboard recommendations. The bioinformatics interest is less directly addressed. The Energy and Industrial Leadership RDIA is consistently and appropriately reflected.

Expected Project Direction

This group would be best suited for an AI-powered cybersecurity web application, such as a full-stack threat intelligence dashboard, a DevSecOps platform with ML-based anomaly detection, or a bioinformatics-informed security risk assessment tool.

Scenario 35 — Embedded IoT and Drone Systems for Environment and Transport

Group: G035 · **Members:** 3 · **Competency:** Moderate · **Interest Pattern:** Diverse · **RDIA:** Sustainable Environment

Top Interests: Internet of Things (IoT), Embedded Systems, Robotics

Top Applications: Sustainability & Environment, Transportation & Mobility, Education

Top Recommended Project: *Energy-aware Multi-Object Detection and Compression in WMSN*

Group Profile

Group G035 is a three-student group with moderate academic performance (average GPA 3.37) whose diverse interests span IoT, embedded systems, robotics, drones, NLP, and data science, with no overlapping interests. The group uniquely features a drone specialist, making it one of the most distinctive profiles in the dataset.

Interest and Application Profile

The three members contribute: IoT and Embedded Systems, Robotics and Drones, and NLP and Data Science. Application priorities include Sustainability and the Environment, Transportation and Mobility, and Education. The sustainability and transport domains are the primary unifying application directions.

Recommendation Outcome

The top recommendation is *Energy-aware Multi-Object Detection and Compression in Wireless Multimedia Sensor Networks*, reflecting the IoT and sustainability alignment. Other top results include drone-based monitoring systems, autonomous vehicle tracking, and smart environmental platforms. The transport and sustainability themes are consistently well-represented.

Alignment Analysis

The alignment is moderate-to-strong. The IoT and drone components are well-captured in the recommendations, while the NLP and data science interests are less prominently reflected. The sustainability and transport application priorities are consistently addressed. The RDIA of Sustainable Environment and Supply of Essential Needs is directly and appropriately reflected.

Expected Project Direction

This group is best positioned for a drone-based or IoT-enabled environmental monitoring project, such as an autonomous drone platform for agricultural or ecological sensing, a smart transport corridor monitoring system, or an NLP-enriched environmental data dashboard.

Scenario 36 — Cross-Domain AI and Cybersecurity Application Development

Group: G036 · **Members:** 3 · **Competency:** Strong · **Interest Pattern:** Diverse · **RDIA:** Energy & Industrial Leadership

Top Interests: Cybersecurity, Artificial Intelligence / Machine Learning, Web Development

Top Applications: Security & Privacy, Healthcare / Medical, Entertainment & Media

Top Recommended Project: *Detecting Malicious PowerShell Scripts using Machine Learning*

Group Profile

Group G036 consists of three students with strong academic performance (average GPA 3.65) whose interests span cybersecurity, AI/ML, web development, software engineering, mobile development, and bioinformatics, with no overlapping interests. This group

represents the most technically heterogeneous configuration among the strong-competency groups.

Interest and Application Profile

The three members contribute: Cybersecurity and Software Engineering, AI/ML and Bioinformatics, and Web Development and Mobile Development. Application priorities include Security and Privacy, Healthcare/Medical, and Entertainment and Media. The security and healthcare domains provide the primary application anchors.

Recommendation Outcome

The top recommendation is *Detecting Malicious PowerShell Scripts using Machine Learning*, reflecting the cybersecurity and AI intersection that bridges the first two members' profiles. Other top results include AI-driven security tools, mobile security applications, and bioinformatics-informed threat modelling systems. The security theme dominates.

Alignment Analysis

The alignment is strong for the security and AI dimensions, but moderate for the web, mobile, and bioinformatics dimensions. The entertainment application preference is underrepresented relative to the healthcare and security themes that dominate the top-10 results. The Energy and Industrial Leadership RDIA is consistently reflected.

Expected Project Direction

This group is best suited for a project that integrates security engineering with AI and cross-platform development, such as a secure mobile health application with AI-driven threat detection, a bioinformatics security risk framework, or a multi-platform cybersecurity management system.